

Constraint-Gated Waveform Design for Low-Excitation Ion Transport

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Abstract

We present a compact computational pipeline for trapped-ion transport waveform design in segmented RF Paul traps. The pipeline maps electrode-basis potentials to moving potential wells, generates smooth voltage waveforms, simulates ion motion, and evaluates residual motional excitation. The implementation uses a transparent one-dimensional model so that relationships between transport speed, waveform smoothness, and excitation are directly observable.

1 Introduction

Segmented trapped-ion architectures require reliable ion transport between trap zones. Transport waveforms must move ions quickly while limiting residual motional excitation. This work develops a small, reproducible simulation pipeline that directly connects waveform design to excitation outcomes.

2 Pipeline

The computational pipeline is:

electrode basis $\rightarrow$ potential well $\rightarrow$ transport path $\rightarrow$ voltage waveform $\rightarrow$ ion motion $\rightarrow$ excitation metric.

3 Transport Model

Ion motion is modeled as a particle in a moving harmonic well:

$$m\ddot{x}(t) = -m\omega^2[x(t) - x_c(t)],$$

where $x_c(t)$ is the waveform-controlled well center.

4 Waveform Design

Smooth transport paths are generated using minimum-jerk interpolation:

$$s(u) = 10u^3 - 15u^4 + 6u^5.$$

This ensures zero velocity and acceleration at endpoints, reducing excitation.

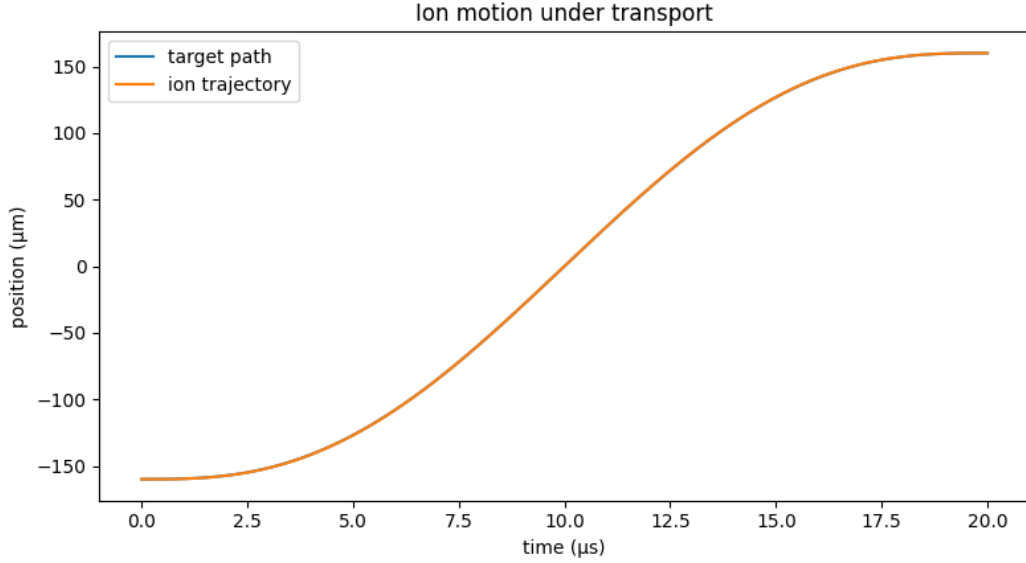


Figure 1: Ion trajectory closely follows the target transport path under minimum-jerk control.

5 Results

5.1 Transport Trajectory

5.2 Path Comparison

5.3 Excitation Scaling

5.4 Frequency-Domain Analysis

6 Discussion

Smooth transport paths suppress resonant excitation by minimizing dynamical forcing. The results demonstrate:

- Orders-of-magnitude reduction in excitation for minimum-jerk paths
- Clear scaling laws linking transport speed and excitation
- Practical control constraints arising from waveform smoothness

7 Conclusion

We presented a compact, interpretable pipeline for ion transport waveform design. The framework directly connects control synthesis to physical outcomes and highlights fundamental tradeoffs between speed and excitation. This approach provides a foundation for more advanced optimal control and experimental implementation.

Code Availability

The project repository is available at:

github.com/thinkthoughts/ion-transport-waveform-pipeline.

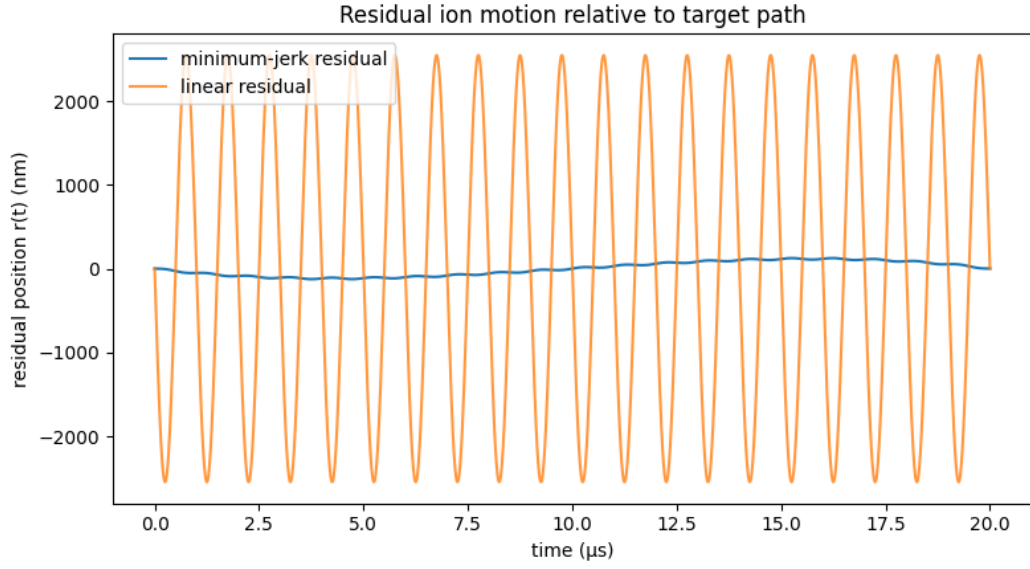


Figure 2: Minimum-jerk transport suppresses oscillatory excitation relative to linear interpolation.

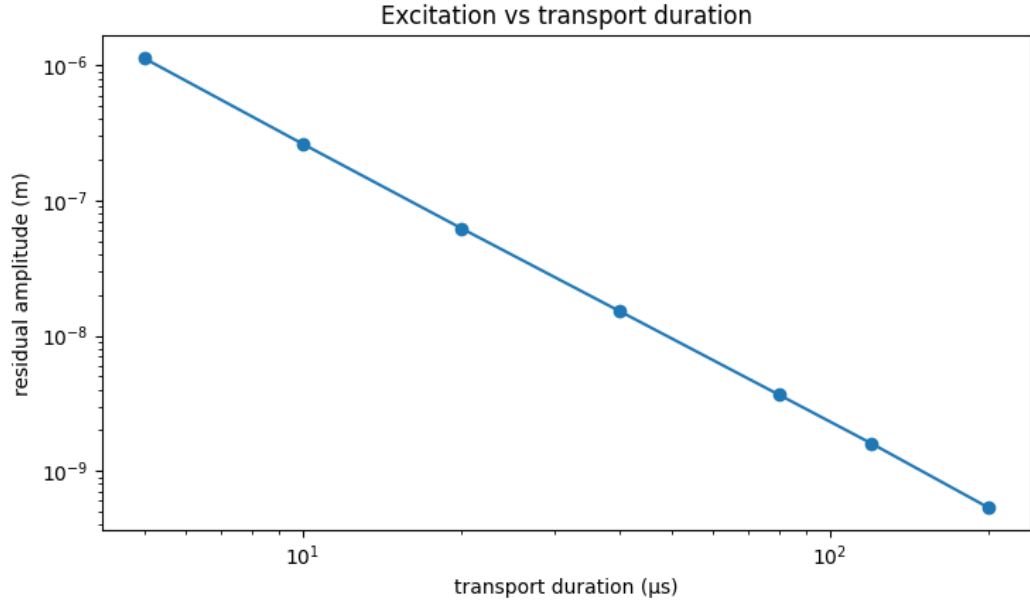


Figure 3: Residual excitation decreases systematically with increasing transport duration.

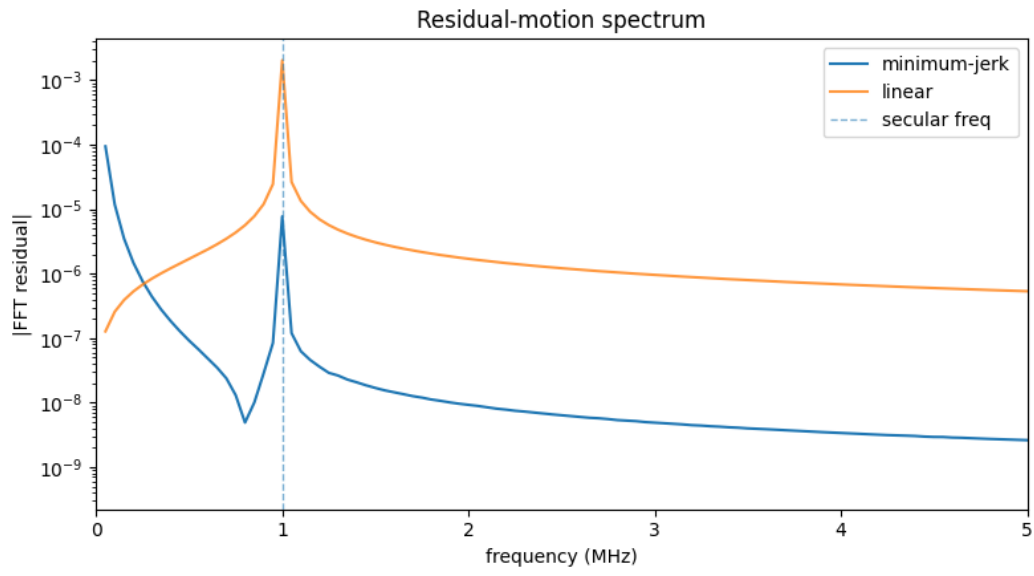


Figure 4: Residual-motion spectrum showing suppression of excitation at the trap secular frequency.